



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 Measurements and Instrumentation

Name of the Faculty member (s): Dr.P.Prem

Innovative Practice Description

- **Unit / Topic:** Unit II / Moving Coil and Moving Iron Instruments
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO4, TLO5, TLO6
- **Activity Chosen:** Demonstration
- **Justification:**

The construction detail of analog meters is taught using the active learning approach of demonstration, Students' involvement in the learning is improved based on this demonstration approach.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about various parts of a typical analog measuring instrument
- ✓ Then I told them to Pair with their neighbours and discuss about the construction of meters for another 1 minute.
- ✓ Finally, I have shown a meter and explained

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	1	1	1	3	-	-	-	1	-	-	-	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 9
	3
Justification for correlation	Adapt tools and techniques to solve engineering problem

- **Images / Screenshot of the practice:**



Reflective Critique:

❖ ***Feedback of practice from students and other stakeholders:***

Students told that the demonstration helped them to understand the concepts easily.

❖ ***Benefit of the practice:***

1. Students understood the concept practically.
2. Encouraged their design based thinking

❖ ***Challenges faced in implementation:***

1. Time utilization for conducting activity.

References:

1. H.S. Kalsi, 'Electronic Instrumentation', McGraw Hill, III Edition 2010.
2. D.V.S. Murthy, 'Transducers and Instrumentation', Prentice Hall of India Pvt Ltd, 2015.
3. David Bell, 'Electronic Instrumentation & Measurements', Oxford University Press, 2013.
4. Martin Reissland, 'Electrical Measurements', New Age International (P) Ltd., Delhi, 2001.
5. Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition, Prentice Hall of India, 2003.

Signature of Faculty Member

HOD



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Course Code & Title: EE3403 Measurements and Instrumentation

Name of the Faculty member (s): Dr.P.Prem

Innovative Practice Description

- **Unit / Topic:** Unit III / Schering Bridge
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO7
- **Activity Chosen:** Demonstration
- **Justification:**

The construction detail of Measuring Bridges is taught using the active learning approach of demonstration, Students' involvement in the learning is improved based on this demonstration approach.

- **Time Allotted for the Activity:** 15 minutes

- **Details of the Implementation:**

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about various arms of a typical measuring bridge
- ✓ Then I told them to Pair with their neighbours and discuss about the construction of Schering bridge for another 1 minute.
- ✓ Finally, I have shown a schering bridge and explained

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	1	-	1	3	-	-	-	-	-	-	-	-	-	-

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 5
	3
Justification for correlation	Adapt tools and techniques to solve engineering problem

- **Images / Screenshot of the practice:**



Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students told that the demonstration helped them to understand the concepts easily.

❖ *Benefit of the practice:*

1. Students understood the concept practically.
2. Encouraged their design based thinking

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.

References:

1. H.S. Kalsi, 'Electronic Instrumentation', McGraw Hill, III Edition 2010.
2. D.V.S. Murthy, 'Transducers and Instrumentation', Prentice Hall of India Pvt Ltd, 2015.
3. David Bell, 'Electronic Instrumentation & Measurements', Oxford University Press, 2013.
4. Martin Reissland, 'Electrical Measurements', New Age International (P) Ltd., Delhi, 2001.
5. Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition, Prentice Hall of India, 2003.

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Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE3403 & Measurements & Instrumentation

Name of the Faculty member (s): Dr.P.Prem

Innovative Practice Description

- **Unit / Topic:** Unit IV / Smart Sensors
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO18
- **Activity Chosen:** Think pair share
- **Justification:**
 - ✓ Since it is a new topic, the activity will make the students explore about the applications of technologies they have already learned.
 - ✓ It enhances the team skills of students
- **Time Allotted for the Activity:** 10 minutes
- **Details of the Implementation:**

The sensors used to measure temperature, pressure, flow, angular velocity were taught in the earlier classes. In this class an introduction about the term “smart” has been given. Then I have posted a question to the students What makes normal sensor, a smart sensor?. Time has been given to students to think and jot down their ideas after discussing with their peers. After discussion, one person from a team presented the requirements of a smart sensor.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	1	1	1	3	-	-	-	-	-	-	-	-	-	-

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO9
	1
Justification for correlation	The students can Function effectively as a team

- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students stated that they understood the concept, and the block diagram given in the reading material is just indicative.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The activity enhanced the participative learning capability of students

- ❖ **Challenges faced in implementation:**

- ✓ The peer discussion deviated from the topic occasionally, so the student needed to be regulated often to create the interest

References:

1. <https://spedellreadingstrategies.weebly.com/think-pair-share.html>
2. <https://www.techtarget.com/iotagenda/definition/smart-sensor>

Signature of Faculty Member

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